

Literature Review: Conditional Generation under Weak Conditioning

Independent Study — Deep Learning

Zhizhe Zhang (zz5070) Dixuan Yang (dy2397)
 NYU Shanghai · Advisor: Prof. Shengjie Wang

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1 Problem and scope

Conditional generative models are normally developed against strong conditions. A text prompt or a class label carries a great deal of information about the output, and the techniques built around these models were tuned in that regime. Much less is known about what they do when the condition is weak: when it shares little information with the target, is buried in noise, and shifts between sessions or subjects. Scalp EEG is an extreme case of all three, which makes it a useful testbed rather than merely a difficult one. This review is organised around one question: how does one tell whether a conditional model uses its condition at all?

2 Decoding continuous hand kinematics from EEG

Table 1 summarises ten studies that regress continuous hand kinematics from low-frequency scalp EEG. Six use WAY-EEG-GAL, the dataset we plan to use, and all are scored by Pearson correlation (PCC).

Study	Task and method	Reported PCC $x/y/z$	Null control
Bradberry 2010	3D center-out reaching, 5 subjects; multi-lag linear decoder on <1 Hz EEG amplitudes	“comparable to intracranial”; falls as variability rises	none reported
Korik 2018	<i>imagined</i> 3D trajectories; mLR on mu/beta/low-gamma band power	band power beats potentials	none reported
Jain & Kumar 2022/23	pre-movement delta EEG (lags 150–350 ms); CNN–LSTM; LOSO added 2023	0.79/0.80/0.63; LOSO 0.76/0.78/0.48	none reported
ESI-GAL 2024	frontoparietal source imaging + rEEGNet	0.790/0.795/0.637 sensor; 0.769/0.777/0.647 source	none reported
M3T-Attention 2026	multi-scale temporal attention transformer, 0.5–12 Hz	0.8816/0.8841/0.8711	none; splits by session within one subject
SAND 2026	spectral + temporal-attention dual branch, five-fold CV	0.9595/0.9534/0.9293	none reported
Antelis 2013	linear regression vs. randomly re-paired data	<i>not above chance</i>	re-pairing
Kobler 2018	visuomotor/oculomotor tracking; hand-at-rest replay control	0.2–0.4 (chance 0.10–0.13; hand at rest 0.33–0.36)	shuffle + behavioural
Agashe 2015	reach-to-grasp hand shape, 15 joints	0.49 (joint angular velocity); grasp type 40% vs. 20%	explicit chance level
Hosseini 2022	phase-based connectivity features (PLV, MSC)	0.43 ± 0.03	two surrogates

Table 1: EEG-to-kinematics studies. The four reporting an explicit null sit below the rule. Bradberry et al. (2010) established the claim the rest of this literature builds on, and one of their secondary results has stayed with us: accuracy fell as movement variability rose. That is consistent with genuine decoding of a signal degraded by variability, but equally with a model that has learned the task-average trajectory, since an average fits worst where trials differ most. Neither that paper nor its successors separate the two readings.

Jain and Kumar (2022, 2023) are closest to our task definition, and theirs is the number we keep returning to: under leave-one-subject-out evaluation their CNN–LSTM reaches roughly 0.76/0.78/0.48, against 0.79/0.80/0.63 within subject. A gap that small across unseen subjects is unexpected for

EEG, which is ordinarily strongly subject-specific. We do not read this as proof of anything; it would be consistent with substantial reliance on subject-invariant task structure or trajectory priors rather than establishing subject-specific neural decoding, and the distinction is testable (§7). M3T-Attention (2026) is our reproduction target because its code is public, though its released script splits by session within a single subject, so its figures are not unseen-subject performance. Read down the table, correlations have risen from roughly 0.5 to 0.96 in four years while the protocol has not changed, and none of the six WAY-EEG-GAL studies reports a chance level, a shuffled-pairing control, or a baseline discarding the EEG input. For scale, intracranial recordings give lower values for comparable targets: 0.50 for position from subdural ECoG (Schalk et al., 2007) and 0.70–0.75 from implanted arrays in macaques (Carmenta et al., 2003). Task, signal and protocol differ, so this is not like-for-like and we do not offer it as a refutation; it does suggest a scalp-EEG correlation of 0.96 is worth checking against a null first.

3 Evaluation pitfalls and null controls

Four studies here did compute a null, and they are the four reporting the lowest correlations. Antelis et al. (2013) compared decoders trained on correct pairings against decoders trained on randomly re-paired EEG and kinematics, and also analysed the metric: PCC is scale-invariant and takes systematically high values on slow, sinusoid-like signals, which describes both low-pass EEG and hand velocity. Under their null the reported correlations were not above chance. Kobler et al. (2018) added a control in which subjects watched a replayed cursor with the hand at rest, which still yielded $r \approx 0.33$ –0.36 for nominal hand velocity against 0.40 during real execution, and that averaging across subjects raised r from 0.40 to 0.83. Agashe et al. (2015) and Hosseini and Shalchyan (2022) report the cleanest positives we found, 0.49 and 0.43 against explicit chance levels, and Combrisson and Jerbi (2015) supply the argument underneath: theoretical chance levels are anticonservative at realistic sample sizes, so a null must be estimated, not assumed. Barachant and Bonnet (2015) offer a contrast: on these same recordings six discrete events were detected at 0.981 AUC, first among 379 Kaggle teams. That problem has been solved to near ceiling under adversarial evaluation; the continuous one never has.

Studies incorporating explicit null controls therefore tend to report substantially lower correlations than those that do not. These are not controlled comparisons, since dataset, task, preprocessing, split, output variable and model capacity all differ across Table 1, so the association is suggestive rather than causal. What it establishes is that the question remains open, and that closing it is inexpensive.

4 Mechanisms of condition neglect

Outside neuroscience this failure has been studied more precisely, and it is generally found by inspecting outputs rather than by any reported metric: Isola et al. (2017) noted in passing that the noise vector given to their pix2pix generator was ignored, and Chefer et al. (2023) named the behaviour in text-to-image diffusion “catastrophic neglect”. It can also be the optimum rather than an optimisation failure.

Chen et al. (2017) argue from a bits-back analysis of the variational bound that when the decoding distribution is powerful enough to model the data without the latent variable, and the approximate posterior does not match the true posterior exactly, the latent variable should be ignored at the optimum, since using it incurs the posterior-approximation gap. Their setting is a learned latent code rather than an external condition, so this transfers by analogy, but the analogy concerns us directly: if a motion prior can model the marginal distribution of reaches on a stereotyped task, a weakly informative EEG condition may be discarded by a correctly trained model rather than in spite of it. Related failures appear as posterior collapse (Bowman et al., 2016).

Shortcut learning describes the general case (Geirhos et al., 2020): rules that do well on i.i.d. test data and fail out of distribution. Lapuschkin et al. (2019) give the sharpest instance, two PASCAL

VOC classifiers of comparable accuracy on the horse class, one reading a photographer’s source tag, which is structurally our concern about ocular artifact. Closest to our situation is Gururangan et al. (2018), whose premise-oblivious classifier, trained on hypothesis sentences alone, reached 67.0% on SNLI against a 34.3% majority baseline; an input-ignoring baseline that scores well is exactly what our template model is meant to be. Where the measurement is weak and the prior strong, the prior fills in structure the data does not support, which Bhadra et al. (2021) make precise by splitting a reconstruction into measurement and null-space components.

The standard metrics do not detect any of this, and the evidence is direct. Ravuri and Vinyals (2019) trained an ImageNet classifier only on samples from a class-conditional generator, using the generator’s own labels, and tested it on real images: BigGAN-deep beats VQ-VAE-2 on FID threefold (11.78 against 38.05) while losing to it by about twelve points on this measure, and over a truncation sweep the two correlate at 0.16. DeVries et al. (2019) make the point by construction, swapping conditionings between matched samples so both marginals are unchanged, whereupon FID is unmoved while their Fréchet Joint Distance responds. This is no criticism of FID (Heusel et al., 2017), which measures distributional similarity; distributional similarity is simply the wrong instrument for asking whether a particular condition was used, and the same applies to the FAD, LSD and DNSMOS scores reported by NeuroSonic.

Condition usage can instead be measured. Van den Oord et al. (2018) connect identification to information: the InfoNCE objective, scoring the true sample against $N - 1$ negatives from the marginal, yields $I \geq \log N - \mathcal{L}$, so identification accuracy among candidates is a lower bound on an information quantity rather than a convenient proxy. McAllester and Stratos (2020) bound what such scores certify, since any distribution-free high-confidence lower bound on mutual information from N samples is at most $2 \ln N + 5$ for $N \geq 50$; that constrains lower bounds only, and does not license the converse claim that a signal’s mutual information is small.

5 Weak neural conditioning, and how adjacent fields evaluate it

Shirakawa et al. (2025) asked whether the perceptually convincing fMRI-to-image reconstructions of Takagi and Nishimoto (2023) carry information specific to the individual trial, evaluating on data built so categories cannot overlap between training and test: performance degrades sharply out of distribution, and they characterise the outputs as classification into trained categories combined with generation by the prior. This is the closest published analogue to the failure we anticipate, in a modality with far better signal quality than scalp EEG, and it was invisible to per-image quality scores. Défossez et al. (2023) decode perceived speech from MEG and EEG across 175 subjects, identifying the correct three-second segment among more than a thousand candidates with up to 41% accuracy: under weak conditioning that field evaluates by identification among candidates, not reconstruction quality. Text-to-motion generation reached the same instrument independently, R-precision (Guo et al., 2022), which measures how often a generated motion retrieves its own text among 32 candidates.

NeuroSonic (Gao et al., 2026) is our closest methodological neighbour, reconstructing speech from scalp EEG by conditional flow matching with x -prediction and reporting gains of up to 26.3% in perceptual quality. Its evaluation rests on FAD, LSD, spectral convergence and DNSMOS, none of which depends on which EEG trial supplied the condition, and it reports no shuffle control, template baseline, condition dropout or guidance analysis. A model that ignored the EEG could therefore obtain favourable scores despite weak trial-specific dependence on it; we do not claim this is what happens, only that the reported evidence does not exclude it. JET (Wang et al., 2026) is more careful, reporting a silhouette score of samples clustered by condition, the only condition-fidelity measure we located here; both build on Lipman et al. (2023).

One result limits the instrument we intend to borrow, and is the strongest counter-evidence we found to our own proposal. Li et al. (2021) re-analysed a widely used EEG classification dataset and

collected new recordings under block and rapid-event designs: high accuracy persisted when class labels were replaced by arbitrary labels varying only by block, and fell to chance under a rapid-event design. Identification on EEG certifies nothing unless the condition has been decorrelated from block, session and time.

6 Synthesis and research gap

These literatures approach the same problem from opposite sides and miss each other. The EEG decoding literature evaluates similarity to a target, which a model that has learned the target’s marginal distribution already satisfies, particularly on a stereotyped task. The generative modelling literature evaluates sample quality and distributional realism, which is independent by construction of which conditioning instance was supplied. Neither, alone, separates a model that has learned a mapping from one that has learned an average.

The central gap is therefore not an absence of stronger generative models but an absence of evaluation protocols that distinguish plausible generation from trial-specific conditional generation. Two adjacent fields arrived independently at the same answer, identification among candidates, and §4 suggests why it is principled rather than conventional. It has not reached EEG-conditioned generation, and in the EEG-to-kinematics literature the corresponding controls are absent from every high-scoring study we found.

7 Dataset, task and first hypotheses

WAY-EEG-GAL (Luciw, Jarocka and Edin, 2014) provides twelve right-handed participants, each performing 328 cued grasp-and-lift trials across ten series, with object weight and surface friction changing unpredictably between trials. It records 32-channel EEG at 500 Hz with no artifact rejection, EMG, force/torque, and a FASTRAK tracking object, index finger, thumb and wrist, all synchronised to within 2 ms, plus sixteen event times we will use for onset alignment. Three properties shape our plan. The usable count is 3,528 trials, not the advertised 3,936, since the non-EEG streams of one series per participant were withheld for the Kaggle competition. The kinematics are nominally 500 Hz, but FASTRAK updates at 120 Hz divided by the number of active sensors, so the effective resolution of our target must be measured before any timing metric is reported. And coverage is sparse with no EOG channel, making ocular contamination a hypothesis to test.

Task. From about one second of EEG preceding movement onset, generate the 3D wrist displacement trajectory of the subsequent reach in a single pass. Baselines under identical splits: a phase-conditioned template ignoring its input, a shuffled-pairing model, an EMG-conditioned positive control, a light EEG regressor, and a reproduced M3T-Attention. A second testbed degrades a clean motion-generation condition, making condition strength an independent variable.

Hypotheses. (1) A template model receiving no EEG reaches per-axis PCC within 0.1 of published values, which would show a high PCC here is by itself insufficient evidence of trial-specific condition usage, though not that PCC is uninformative in general. (2) A decoder trained on randomly repaired data is not separable from a correctly paired one by PCC but is by mean displacement error, with 50–100 retrained shuffled models giving the null under both, and (3) the five baselines rank differently under the two. (4) For the reproduced M3T, LOSO performance falls within 0.05 PCC of within-subject performance, consistent with substantial reliance on subject-invariant task structure, though not by itself evidence that no subject-specific information is used, since subject-invariant neural components are also possible. (5) Removing Fp1, Fp2, F7 and F8 does not push the model below the template level; a collapse to chance would indicate predictive information concentrated in frontal channels consistent with ocular contamination, though attributing it to artifact would need EOG regression and channel-matched controls. (6) Following Li et al. (2021), a trial-identification measure may acquire diagonal structure from the clock alone and none of the controls above would catch it, so we add a timestamp-only baseline and use across-series pairs only.

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